

“PAPER_{0:D3}” -f [int]# PAPER #153 — UQFF Morris-Thorne Wormhole Geodesics: UQFF Metric Integration

Title: UQFF Star-Magic Morris-Thorne Wormhole Metric — fTRZ Throat Geometry and Geodesic Structure in the MUGE 12-Term Resonance Framework

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\text{fTRZ}=0.1$)

Date: March 2026

Domain: §2.2 MUGE Compression Cycle 3 (07b7f7a6)

Source Thread: `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt`

UQFF Mode: Superconductive Resonance (exotic geometry)

Validator: `CondensedPhysics2.py v2.1.0` (exotic geometry module)

Cross-links: PAPER_152 (cosmological baseline), PAPER_154 (Navier-Stokes jets), PAPER_155 (SM limiting case)

Abstract

The Morris-Thorne traversable wormhole metric provides one of general relativity’s most physically transparent windows into exotic spacetime topology. In the UQFF Star-Magic framework, the wormhole throat is identified as a superconductive manifold (SCm) junction — a localised maximum of the Ug_4 vacuum concentration field where the topological resonance factor $f_{\text{TRZ}} = 0.1$ governs the suppression of the shape function $b(r)$ relative to the FLRW background. This paper derives the UQFF-modified Morris-Thorne line element, demonstrates that the throat radius is set by the SCm-vacuum equilibrium condition $\rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 = \rho_{\text{vac}} \cdot c^2$, computes the UQFF geodesic equations through the throat, and shows that the fTRZ term contributes a physically meaningful correction to the exotic matter requirement: the UQFF framework reduces the required negative energy density by a factor of $f_{\text{TRZ}} = 0.1$ via the vacuum concentration field Ug_4 . The paper also connects the wormhole throat MUGE resonance value to the $\text{fTRZ} = 0.1$ contribution in the 12-term master equation.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Morris-Thorne Wormhole Primer

1.1 Standard MT Metric

The traversable wormhole line element (Morris & Thorne 1988) in spherical coordinates (t, r, θ, ϕ) :

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \frac{dr^2}{1 - b(r)/r} + r^2d\Omega^2$$

where: - $\Phi(r)$ = redshift function (zero for zero-tidal-force wormhole: $\Phi = 0$) - $b(r)$ = shape function satisfying $b(r_0) = r_0$ at the throat r_0 - Flare-out condition: $b'(r_0) < 1$

The exotic matter requirement from the Einstein field equations:

$$\rho_{exotic} = -\frac{c^2}{8\pi G} \frac{b'}{r^2} < 0$$

For a traversable wormhole of throat radius r_0 , the required negative energy density is:

$$|\rho_{exotic}| \sim \frac{c^2}{8\pi G r_0^2}$$

For $r_0 = 1$ m: $|\rho_{exotic}| \sim 10^{25}$ J/m³ — far exceeding any known laboratory energy density.

1.2 The UQFF Resolution

In UQFF, the SCm (superconducting manifold) provides a physical realization of the exotic energy density through the vacuum concentration field Ug4i. The SCm energy density:

$$\rho_{SCm} \cdot v_{SCm}^2 = 10^{15} \times (10^8)^2 = 10^{31} \text{ J/m}^3$$

This exceeds the exotic requirement for macroscopic wormholes ($r_0 \sim 1$ m) by 6 orders of magnitude, suggesting UQFF naturally provides the exotic matter source.

2. UQFF-Modified Morris-Thorne Metric

2.1 The UQFF fTRZ Correction

In the UQFF framework, the topological resonance zone (TRZ) modifies the shape function:

$$b_{UQFF}(r) = b_0(r) \cdot (1 - f_{TRZ}) + f_{TRZ} \cdot r_0 \cdot e^{-\kappa(r-r_0)}$$

where $b_0(r) = r_0^2/r$ is the simplest MT shape function. Substituting $f_{TRZ} = 0.1$:

$$b_{UQFF}(r) = 0.9 \cdot \frac{r_0^2}{r} + 0.1 \cdot r_0 \cdot e^{-\kappa(r-r_0)}$$

At throat ($r = r_0$, $\kappa(r - r_0) = 0$): $b_{UQFF}(r_0) = 0.9r_0 + 0.1r_0 = r_0$?

The $f_{TRZ} = 0.1$ term adds an exponentially-decaying vacuum concentration component from the Ug4i field. This modifies the flare-out condition:

$$b'_{UQFF}(r_0) = -0.9\frac{r_0^2}{r_0^2} - 0.1\kappa r_0 = -0.9 - 0.1\kappa r_0 < 1$$

The flare-out condition is automatically satisfied for any $r_0 > 0$ since the left side is always negative.

2.2 The UQFF MT Line Element

$$\begin{aligned} ds_{UQFF}^2 &= -c^2 dt^2 + \frac{dr^2}{1 - \frac{0.9r_0^2/r + 0.1r_0 e^{-\kappa(r-r_0)}}{r}} + r^2 d\Omega^2 \\ &= -c^2 dt^2 + \frac{dr^2}{1 - \frac{0.9r_0^2}{r^2} - \frac{0.1r_0}{r} e^{-\kappa(r-r_0)}} + r^2 d\Omega^2 \end{aligned}$$

This is the **UQFF Morris-Thorne** metric — the shape function has two components:
1. Standard flare (0.9 factor, GR-like) 2. UQFF Ug4i vacuum concentration (0.1 factor, exponentially localised at throat)

2.3 SCm Throat Equilibrium Condition

The throat radius r_0 is determined by the SCm-vacuum equilibrium:

$$\begin{aligned} \rho_{SCm} \cdot v_{SCm}^2 &= \frac{c^2}{8\pi G r_0^2} \\ r_0^2 &= \frac{c^2}{8\pi G \rho_{SCm} v_{SCm}^2} = \frac{(3 \times 10^8)^2}{8\pi \times 6.67 \times 10^{-11} \times 10^{15} \times (10^8)^2} \\ r_0^2 &= \frac{9 \times 10^{16}}{8\pi \times 6.67 \times 10^{-11} \times 10^{31}} = \frac{9 \times 10^{16}}{1.676 \times 10^{22}} = 5.37 \times 10^{-6} \text{ m}^2 \\ r_0 &= \sqrt{5.37 \times 10^{-6}} \approx 2.32 \times 10^{-3} \text{ m} = 2.32 \text{ mm} \end{aligned}$$

The UQFF MUGE framework predicts **a natural wormhole throat radius of ~2.3 mm** set by the SCm energy density parameters. This is the minimum macroscopic wormhole size consistent with UQFF vacuum physics.

3. Geodesic Equations Through the UQFF Throat

3.1 Radial Geodesics (Equatorial Plane)

For a timelike geodesic with $\dot{t} = 1$ (zero tidal force: $\Phi = 0$, so $e^{2\Phi} = 1$), the geodesic equation for the radial coordinate:

$$\ddot{r} + \Gamma_{rr}^r \dot{r}^2 = 0$$

where the Christoffel symbol with the UQFF shape function:

$$\Gamma_{rr}^r = \frac{b'_{UQFF}r - b_{UQFF}}{2r^2(1 - b_{UQFF}/r)}$$

At the throat ($b_{UQFF}(r_0) = r_0$, $b'_{UQFF}(r_0) = -0.9 - 0.1\kappa r_0$):

$$\Gamma_{rr}^r|_{r_0} = \frac{(-0.9 - 0.1\kappa r_0)r_0 - r_0}{2r_0^2 \cdot 0} \rightarrow \text{indeterminate}$$

The throat is a coordinate singularity in r — standard MT behavior. In the UQFF framework, the passage through the throat is smooth because the fTRZ exponential term regularizes the shape function:

$$\lim_{r \rightarrow r_0} (1 - b_{UQFF}/r) = \lim_{r \rightarrow r_0} \left(1 - 0.9 \frac{r_0^2}{r^2} - 0.1 \frac{r_0}{r} \right) = 1 - 0.9 - 0.1 = 0$$

The zero is first-order, confirming traversability. A traveler crossing the throat experiences:

$$\tau_{transit} \approx \frac{r_0}{v_{traveler}} \approx \frac{2.32 \times 10^{-3}}{c} \approx 7.7 \times 10^{-12} \text{ s}$$

At the UQFF SCm throat, transit time is sub-picosecond — the wormhole is instantaneous at human scales.

3.2 MUGE Gravity at the Throat

The MUGE 12-term resonance value at the wormhole throat ($r = r_0 = 2.32 \text{ mm}$, $B = B_{SCm}$):

The dominant term at the ultra-dense throat is a_{aether_res} (SCm velocity dominates):

$$a_{aether_res,throat} = \gamma \cdot \rho_{SCm} \cdot v_{SCm} \cdot c = 5 \times 10^{-5} \times 10^{15} \times 10^8 \times 3 \times 10^8 = 1.5 \times 10^{27} \text{ m/s}^2$$

This is comparable to the Sgr A* MUGE value (4.105×10^{29}) — consistent with the extreme spacetime curvature at a macroscopic wormhole throat.

The fTRZ contribution at the throat directly:

$$f_{TRZ,throat} = 0.1 \text{ m/s}^2 \text{ reference contribution}$$

The fTRZ = 0.1 is the normalised UQFF throat-resonance unit — setting the scale for how the TRZ contribution per unit volume relates to the total MUGE field.

4. Exotic Matter Reduction via UQFF

4.1 Standard Exotic Matter Requirement

For a wormhole with $r_0 = 2.32 \text{ mm}$:

$$|\rho_{exotic,GR}| = \frac{c^2}{8\pi G r_0^2} = \frac{9 \times 10^{16}}{8\pi \times 6.67 \times 10^{-11} \times 5.37 \times 10^{-6}} \approx 10^{31} \text{ J/m}^3$$

4.2 UQFF Reduced Exotic Requirement

In UQFF, the Ug4i vacuum concentration field contributes positive effective energy density that partially cancels the exotic requirement:

$$\rho_{exotic,UQFF} = \rho_{exotic,GR} \cdot (1 - f_{TRZ}) \cdot e^{-\kappa t}$$

With $f_{TRZ} = 0.1$ and $e^{-\kappa t} \approx 0.08$ (at cosmological time):

$$\rho_{exotic,UQFF} = 10^{31} \times 0.9 \times 0.08 \approx 7.2 \times 10^{29} \text{ J/m}^3$$

The **UQFF framework reduces the exotic matter requirement by ~93%** relative to GR alone, with the remaining $7.2 \times 10^{29} \text{ J/m}^3$ provided by the SCm energy density ($\rho_{SCm} \cdot v_{SCm} = 10^{31} \text{ J/m}^3 > \text{required}$).

This demonstrates that **UQFF wormholes are energetically self-consistent** — the SCm density exceeds the reduced exotic requirement by more than $10\times$.

5. Connection to MUGE 12-Term Master Equation

The fTRZ = 0.1 contribution in the 12-term master equation:

$$g(r, t) = \dots + f_{TRZ} = \dots + 0.1$$

is precisely the normalised wormhole throat resonance contribution — the dimensionless scale factor connecting the metric topology (non-trivial π_1 of the spacetime manifold) to the MUGE gravity sum. In regimes where the other 11 terms dominate, f_{TRZ} is a small correction. At the wormhole throat, f_{TRZ} becomes the defining topology-gravity coupling.

The topological interpretation: - $f_{TRZ} = 0.1$? the throat contributes 10% of the MUGE field energy as a topology term - $(1 - f_{TRZ}) = 0.9$? 90% of the MUGE field is resonance-mediated (non-topological) - This 90/10 split mirrors the $O_{\text{?}}/O_{\text{m}}$ ratio in Λ CDM ($0.685/0.315 \sim 2.2$) at the order-of-magnitude level

6. Physical Predictions

6.1 Observable Signatures

Prediction	UQFF Value	Observable
Throat radius	$r_0 = 2.32 \text{ mm}$	— (hypothetical)
Throat MUGE field	$\sim 1.5 \times 10^{27} \text{ m/s}^2$	— (gravitational lensing pattern)
Transit time at light speed	$\sim 7.7 \text{ ps}$	—
Exotic matter reduction	93% vs GR	—
f_{TRZ} geometry signature	10% topology coupling	Lensing ring asymmetry
Shape function decay rate	$\text{?} = 5 \times 10^{-4} / \text{day}$? spatial	exponential falloff

6.2 Connection to Einstein Ring Lensing

The Rings of Relativity system (PAPER_151) is an Einstein ring — a near-zero-tidal-force perfect alignment geometry analogous to the zero-tidal-force condition in MT wormholes ($\Phi = 0$). The UQFF connection:

The lensing ring geometry satisfies the same mathematical condition as the MT metric ($\$^{2\Box} = \$ \text{const}$) when the MUGE field at the lens plane activates the f_{TRZ} term. The Rings of Relativity MUGE value $g = 5.005 \times 10^{25} \text{ m/s}^2$ can be interpreted as the UQFF field of a “macroscopic lensing throat” at the Einstein ring — a topologically non-trivial gravitational geometry where the lens bends light through exactly 360° (complete ring), the UQFF analogue of a wormhole mouth.

7. Key Results Summary

Quantity	Value	Units
UQFF wormhole throat radius	2.32 mm	m
Throat equilibrium condition	$\text{?}_{SCm} \cdot v^2_{SCm} = c^2 / (8pGr0^2)$	—
f_{TRZ} shape function contribution	10% (exponentially localised)	—

Quantity	Value	Units
UQFF exotic matter requirement	7.2×10^{29}	J/m ³
SCm local energy density	10^{31}	J/m ³
Exotic matter self-sufficiency	SCm > required by 14×	—
Transit time (at c)	7.7 ps	s
MUGE at throat	$\sim 1.5 \times 10^{27}$	m/s ²
fTRZ topology/metric coupling	10% / 90% topology/resonance	—

8. Conclusions

1. The UQFF Morris-Thorne wormhole metric incorporates $f_{\text{TRZ}} = 0.1$ as a shape function correction that adds a physically motivated exponentially-localised vacuum concentration component to the standard MT shape function.
2. The equilibrium throat radius predicted by UQFF is $r_0 \sim 2.32$ mm — set entirely by the SCm energy density parameters (ρ_{SCm} , v_{SCm}) with no free parameters.
3. The UQFF framework reduces the exotic matter requirement by 93% relative to GR, and the remainder is fully supplied by the SCm energy density.
4. The $f_{\text{TRZ}} = 0.1$ term in the MUGE master equation has a direct topological interpretation as the 10% coupling between spacetime topology and the resonance gravity field.
5. The Rings of Relativity Einstein ring (PAPER_151) represents the UQFF cosmological analogue of a wormhole throat, and the $g_{\text{MUGE}} = 5.005 \times 10^{25}$ m/s² result is the far-field signature of this topology.

UQFF computed: Canonical UQFF buoyancy parameter $U_{\text{bi}} = \rho \times [\text{SSq}] \times GM/r^2 = 5.0\text{e-}4 \times 0.57 \times 6.67\text{e-}11 \times M/r^2$; for solar parameters: $U_{\text{bi,Sun}} = 5.7\text{e-}4 \times 6.67\text{e-}11 \times 1.99\text{e}30 / (6.96\text{e}8)^2 = 1.47\text{e+}2$ m/s².

References

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- SOURCE4 namespace, MAIN_1_CoAnQi.cpp lines 25623–26026
- grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt — Thread 07b7f7a6 .Groups[1].Value — UQFF Morris-Thorne Wormhole Geodesics: UQFF Metric Integration

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where $b_0(r) = r_0^2/r$ is the simplest MT shape function. Substituting $f_{TRZ} = 0.1$:

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The fTRZ = 0.1 term adds an exponentially-decaying vacuum concentration component from the Ug4i field. This modifies the flare-out condition:

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The flare-out condition is automatically satisfied for any $r_0 > 0$ since the left side is always negative.

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This is the **UQFF Morris-Thorne** metric — the shape function has two components: 1. Standard flare (0.9 factor, GR-like) 2. UQFF Ug4i vacuum concentration (0.1 factor, exponentially localised at throat)

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The throat radius r_0 is determined by the SCm-vacuum equilibrium:

$$\rho_{SCm} \cdot v_{SCm}^2 = \frac{c^2}{8\pi G r_0^2}$$

$$r_0^2 = \frac{c^2}{8\pi G \rho_{SCm} v_{SCm}^2} = \frac{(3 \times 10^8)^2}{8\pi \times 6.67 \times 10^{-11} \times 10^{15} \times (10^8)^2}$$

$$r_0^2 = \frac{9 \times 10^{16}}{8\pi \times 6.67 \times 10^{-11} \times 10^{31}} = \frac{9 \times 10^{16}}{1.676 \times 10^{22}} = 5.37 \times 10^{-6} \text{ m}^2$$

$$r_0 = \sqrt{5.37 \times 10^{-6}} \approx 2.32 \times 10^{-3} \text{ m} = 2.32 \text{ mm}$$

The UQFF MUGE framework predicts **a natural wormhole throat radius of ~2.3 mm** set by the SCm energy density parameters. This is the minimum macroscopic wormhole size consistent with UQFF vacuum physics.

3. Geodesic Equations Through the UQFF Throat

3.1 Radial Geodesics (Equatorial Plane)

For a timelike geodesic with $\dot{t} = 1$ (zero tidal force: $\Phi = 0$, so $e^{2\Phi} = 1$), the geodesic equation for the radial coordinate:

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where the Christoffel symbol with the UQFF shape function:

$$\Gamma_{rr}^r = \frac{b'_{UQFF}r - b_{UQFF}}{2r^2(1 - b_{UQFF}/r)}$$

At the throat ($b_{UQFF}(r_0) = r_0$, $b'_{UQFF}(r_0) = -0.9 - 0.1\kappa r_0$):

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The throat is a coordinate singularity in r — standard MT behavior. In the UQFF framework, the passage through the throat is smooth because the fTRZ exponential term regularizes the shape function:

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The MUGE 12-term resonance value at the wormhole throat ($r = r_0 = 2.32 \text{ mm}$, $B = B_{SCm}$):

The dominant term at the ultra-dense throat is aaether_res (SCm velocity dominates):

$$a_{aether_res,throat} = \gamma \cdot \rho_{SCm} \cdot v_{SCm} \cdot c = 5 \times 10^{-5} \times 10^{15} \times 10^8 \times 3 \times 10^8 = 1.5 \times 10^{27} \text{ m/s}^2$$

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4. Exotic Matter Reduction via UQFF

4.1 Standard Exotic Matter Requirement

For a wormhole with $r_0 = 2.32$ mm:

$$|\rho_{exotic,GR}| = \frac{c^2}{8\pi G r_0^2} = \frac{9 \times 10^{16}}{8\pi \times 6.67 \times 10^{-11} \times 5.37 \times 10^{-6}} \approx 10^{31} \text{ J/m}^3$$

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With $f_{TRZ} = 0.1$ and $e^{-\kappa t} \approx 0.08$ (at cosmological time):

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5. Connection to MUGE 12-Term Master Equation

The $f_{TRZ} = 0.1$ contribution in the 12-term master equation:

$$g(r, t) = \dots + f_{TRZ} = \dots + 0.1$$

is precisely the normalised wormhole throat resonance contribution — the dimensionless scale factor connecting the metric topology (non-trivial π_1 of the spacetime manifold) to the MUGE gravity sum. In regimes where the other 11 terms dominate, f_{TRZ} is a small correction. At the wormhole throat, f_{TRZ} becomes the defining topology-gravity coupling.

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6.1 Observable Signatures

Prediction	UQFF Value	Observable
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Transit time at light speed	$\sim 7.7 \text{ ps}$	—
Exotic matter reduction	93% vs GR	—
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6.2 Connection to Einstein Ring Lensing

The Rings of Relativity system (PAPER_151) is an Einstein ring — a near-zero-tidal-force perfect alignment geometry analogous to the zero-tidal-force condition in MT wormholes ($\Phi = 0$). The UQFF connection:

The lensing ring geometry satisfies the same mathematical condition as the MT metric ($\$^{2\Box} = \$ \text{const}$) when the MUGE field at the lens plane activates the fTRZ term. The Rings of Relativity MUGE value $g = 5.005 \times 10^{25} \text{ m/s}^2$ can be interpreted as the UQFF field of a “macroscopic lensing throat” at the Einstein ring — a topologically non-trivial gravitational geometry where the lens bends light through exactly 360° (complete ring), the UQFF analogue of a wormhole mouth.

7. Key Results Summary

Quantity	Value	Units
UQFF wormhole throat radius	2.32 mm	m
Throat equilibrium condition	$\gamma_{\text{SCm}} \cdot v^2_{\text{SCm}} = c^2 / (8pGr_0^2)$	—
fTRZ shape function contribution	10% (exponentially localised)	—
UQFF exotic matter requirement	7.2×10^{29}	J/m^3
SCm local energy density	10^{31}	J/m^3
Exotic matter self-sufficiency	$\text{SCm} > \text{required by } 14\times$	—
Transit time (at c)	7.7 ps	s
MUGE at throat	$\sim 1.5 \times 10^{27}$	m/s^2
fTRZ topology/metric coupling	10% / 90% topology/resonance	—

8. Conclusions

1. The UQFF Morris-Thorne wormhole metric incorporates $f_{\text{TRZ}} = 0.1$ as a shape function correction that adds a physically motivated exponentially-localised vacuum concentration component to the standard MT shape function.

2. The equilibrium throat radius predicted by UQFF is $r_0 \sim 2.32$ mm — set entirely by the SCm energy density parameters (ρ_{SCm} , v_{SCm}) with no free parameters.
3. The UQFF framework reduces the exotic matter requirement by 93% relative to GR, and the remainder is fully supplied by the SCm energy density.
4. The $f_{TRZ} = 0.1$ term in the MUGE master equation has a direct topological interpretation as the 10% coupling between spacetime topology and the resonance gravity field.
5. The Rings of Relativity Einstein ring (PAPER_151) represents the UQFF cosmological analogue of a wormhole throat, and the $g_{MUGE} = 5.005 \times 10^{25}$ m/s² result is the far-field signature of this topology.

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \rho_{SCm} \times GM/r^2 = 5.0e-4 \times 0.57 \times 6.67e-11 \times M/r^2$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \times 6.67e-11 \times 1.99e30 / (6.96e8)^2 = 1.47e+2$ m/s².

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